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BlenDaViz, scientific visualization library for Blender in Python

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Summary

BlenDaViz is a 3d plotting library for the visualization tool Blender, written in Python. It is able to visualize line plots, scatter plots, surfaces, iso-surfaces, glyphs (arrows) and streamlines from Numpy arrays. The user can load BlenDaViz within a Blender Python console as any other library. Its design is close to other plotting libraries, especially Matplotlib, which means that every part of a plot is treated as an object that can be changed when updating the plot with new parameters or data. Compared to other 3d visualization software, we achieve higher image quality, thanks to Blender's ray tracing capabilities.

Statement of need

In scientific fields, like structure mechanics, computational fluid dynamics, astrophysics, or molecular dynamics, it is common to use 3d visualization tools like Paraview (Ahrens et al., 2005), Vapor (Gruchalla et al., 2009), Mayavi or Visit. They are capable to load a variety of data formats and visualize different aspects of the data using filters. However, they often lack the image quality needed that would make understanding the data easy. Their limited shaders can create optical illusions where the depth of the geometry is not clear, or the colors are off due to unrealistic image generation.

Realistically rendered or ray-traced images, like those generated by 3d modelling tools like Blender, not only improve the image quality, but make it easier to understand the three-dimensional structure and colors. For instance, contextually-implied scattered light can have a significant impact on the perceived color of objects, as exemplified by *The Dress* controversy (Chetverikov & Ivanchei, 2016). Using lighting, shadow, depth and scattering can be used to generate photorealistic images where the interpretation is made easier.

State of the field

So far Blender has seen only limited use in 3d data visualization. Bioblender has been developed specifically for biological data and requires significant Blender knowledge. Similarly, suggestions to use Blender in Astrophysics (Kent, 2015) require the generation of the geometry by hand. High quality plots by the Oak Ridge National Laboratory shows that to achieve high quality a collaboration between Blender experts and scientists is sometimes required.

Attempts at creating libraries were made with AstroBlend (Naiman, 2016) and BlenderSciViz, but their applications are relatively limited and they have not been maintained in years. Perhaps BVtkNodes and SciBlend (Marín et al., 2025) stand out, as they are able to load VTK data and apply filters on it.

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With BlenDaViz (Simon Candelaresi & Berg Smiet, 2025) we give the users all the rendering and ray tracing capabilities of Blender, combined with the ease of use and modularity we know from libraries like Matplotlib (Caswell et al., 2020; Hunter, 2007). It is general purpose, thanks to the implementation of some of the most used 3d plotting methods.

Software design

With BlenDaViz we keep the design very close to Matplotlib. The user supplies the input data in the form of Numpy arrays and the resulting plots are objects that can be freely modified. These objects contain members, such as the data and parameters (e.g. color), and methods, like the plotting function. That way, the user can modify their plots and replot them. An example is shown here where we generate a simple line plot, change the data of the resulting plot object and replot it:

```
import blendaviz as blt
import numpy as np

# Generate the data.
y = np.linspace(0, 6*np.pi, 400)
x = 2*np.cos(y)
z = 2*np.sin(y)

# Generate the line plot.
pl = blt.plot(x, y, z, radius=0.5)

# Change the data.
pl.x = 2*pl.x

# Replot the data.
pl.plot()
```

Each of our plotting functions follows a similar pipeline. As the plotting function is called, the input is checked for consistency. Then any existing geometry or textures for this plot are deleted. Depending on the plotting function, different kind of meshes are generated (lines, surfaces, arrows, etc.). To finalize the plot object a texture is added. Geometry and texture are then part of the plotting object and can, in principle, be accessed. After the geometry generation we add lights and a camera, if not already present.

An example is shown in Figure 1 where we plot a two-dimensional array as a surface with the values used as z-coordinates as colors. To highlight its geometry we added a plane with a checker board texture in the background and readjusted the camera. Such manual adjustments can be done directly in Blender and can greatly improve the plot quality.

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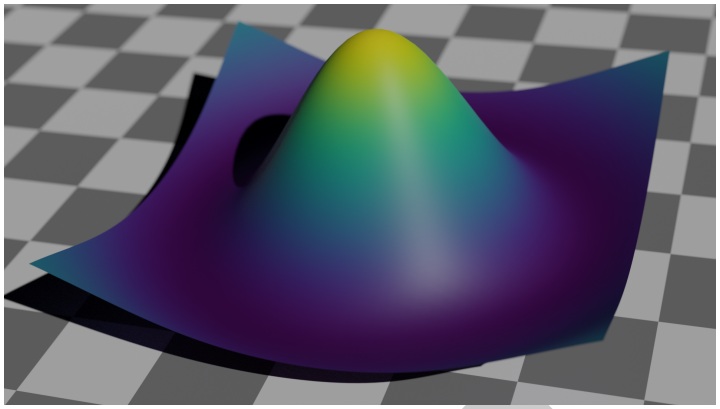


Figure 1: Mesh plot of a Gaussian using a colormap.

Handling of time

We make handling time dependent data easy in BlenDaViz. All that is required is that the data arrays contain one more dimension. For instance, if we want to plot the position of n particles at p different times, our arrays x, y and z would have the shape $[n, p]$. Parameters that take the same shape, like colors, follow the same structure.

BlenDaViz recognizes the data and input parameters as time dependent. It then adds the plotting method of this plot object to the `bpy.app.handlers.frame_change_pre` handler of Blender. That way, any change in the time via e.g. the slider, results in an update of all present plots.

Research impact statement

In (S. Candelaresi et al., 2021) it was used to visualize the vortex field of a turbulent finite difference simulation using the Pencil Code. Later (Israeli et al., 2023) and (S. Candelaresi & Beck, 2023) used it to visualize magnetic field configurations obtained from simulations. The former also uses BlenDaViz to present particle trajectories.

AI usage disclosure

We used Claude Sonnet 4.6 for some code debugging and to help write the majority of tests. All changed by Claude were verified by hand.

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